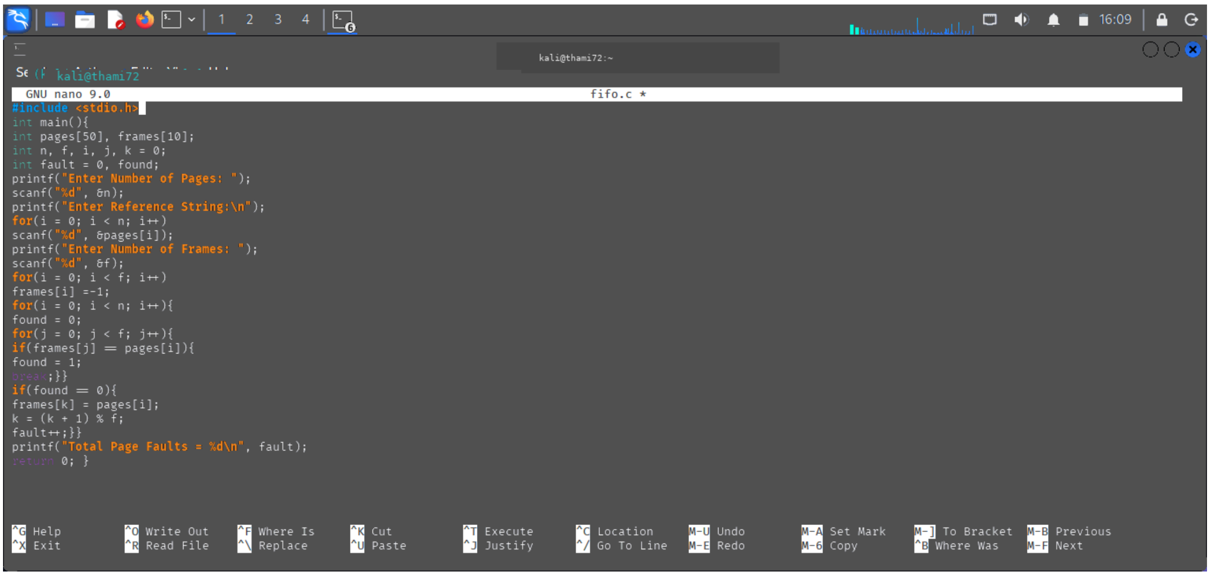
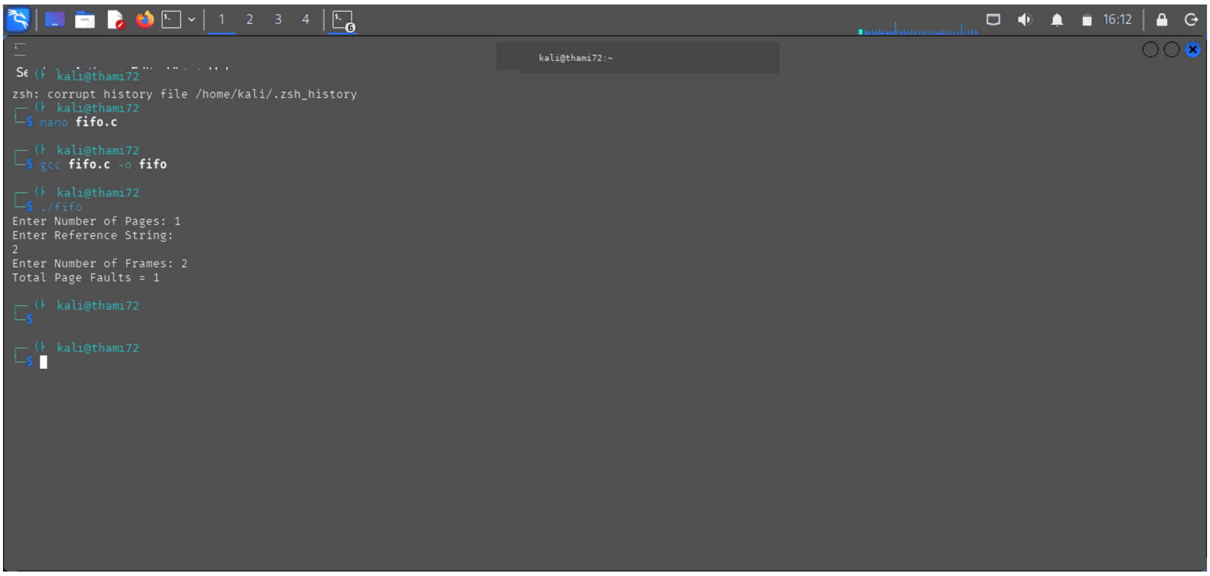
C PROGRAMMING : FIFO



```
GNU nano 9.0                    fifo.c *
#include <stdio.h>
int main(){
int pages[50], frames[10];
int n, f, i, j, k = 0;
int fault = 0, found;
printf("Enter Number of Pages: ");
scanf("%d", &n);
printf("Enter Reference String:\n");
for(i = 0; i < n; i++)
scanf("%d", &pages[i]);
printf("Enter Number of Frames: ");
scanf("%d", &f);
for(i = 0; i < f; i++)
frames[i] =-1;
for(i = 0; i < n; i++){
found = 0;
for(j = 0; j < f; j++){
if(frames[j] == pages[i]){
found = 1;
break;}}
if(found == 0){
frames[k] = pages[i];
k = (k + 1) % f;
fault++;}}
printf("Total Page Faults = %d\n", fault);
return 0; }
```

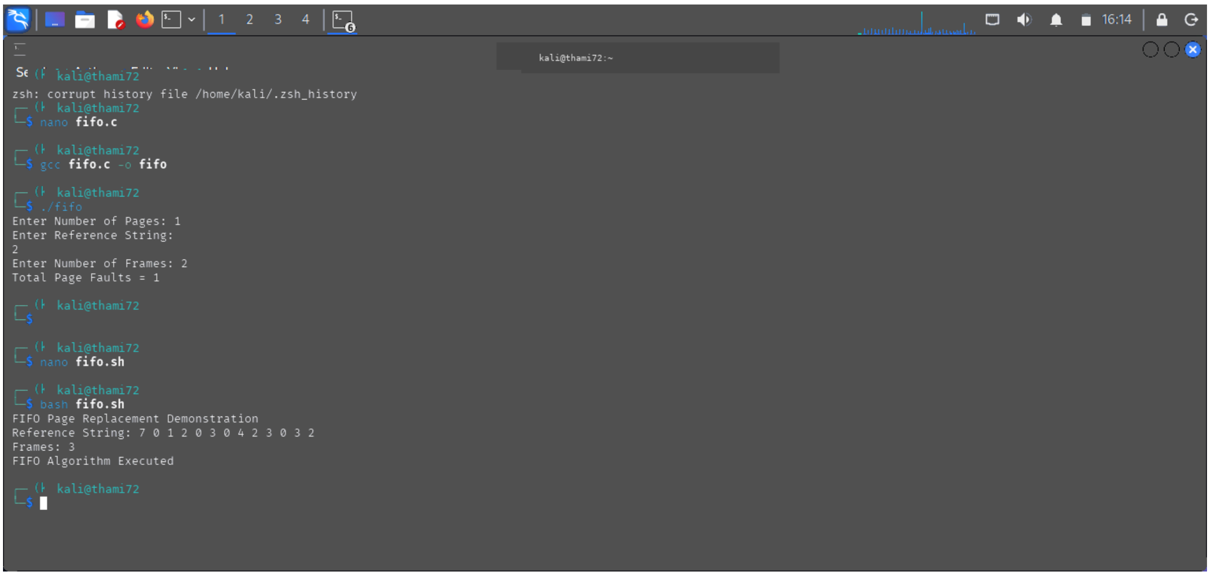
OUTPUT :



```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72
$ nano fifo.c

kali@thami72
$ gcc fifo.c -o fifo

kali@thami72
$ ./fifo
Enter Number of Pages: 1
Enter Reference String:
2
Enter Number of Frames: 2
Total Page Faults = 1

kali@thami72
$

kali@thami72
$
```

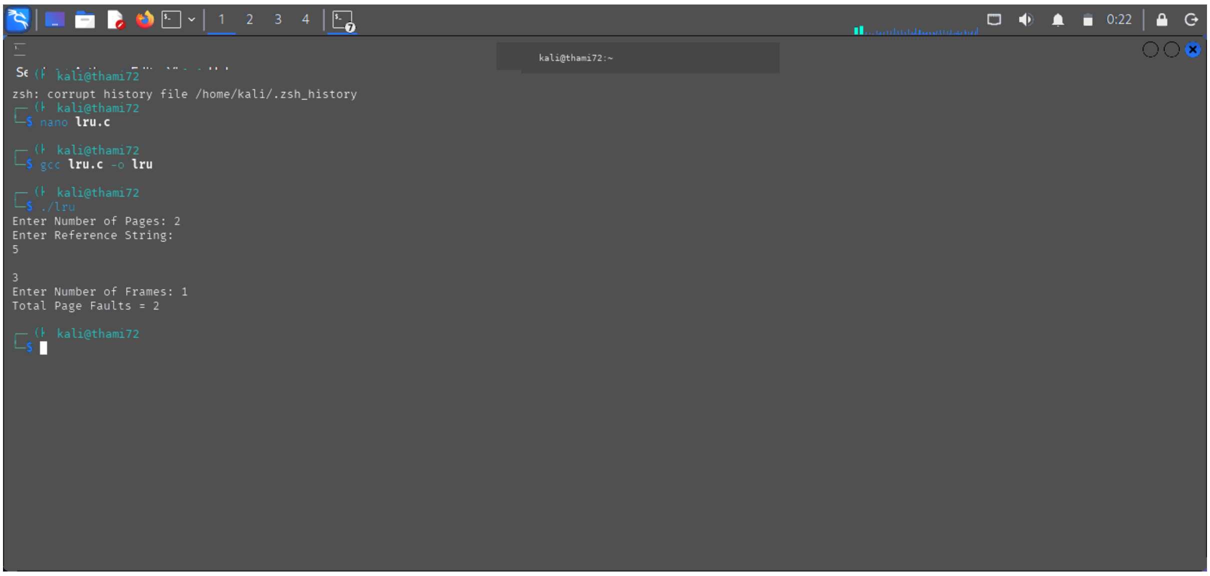
SHELL PROGRAMMING : FIFO

OUTPUT :



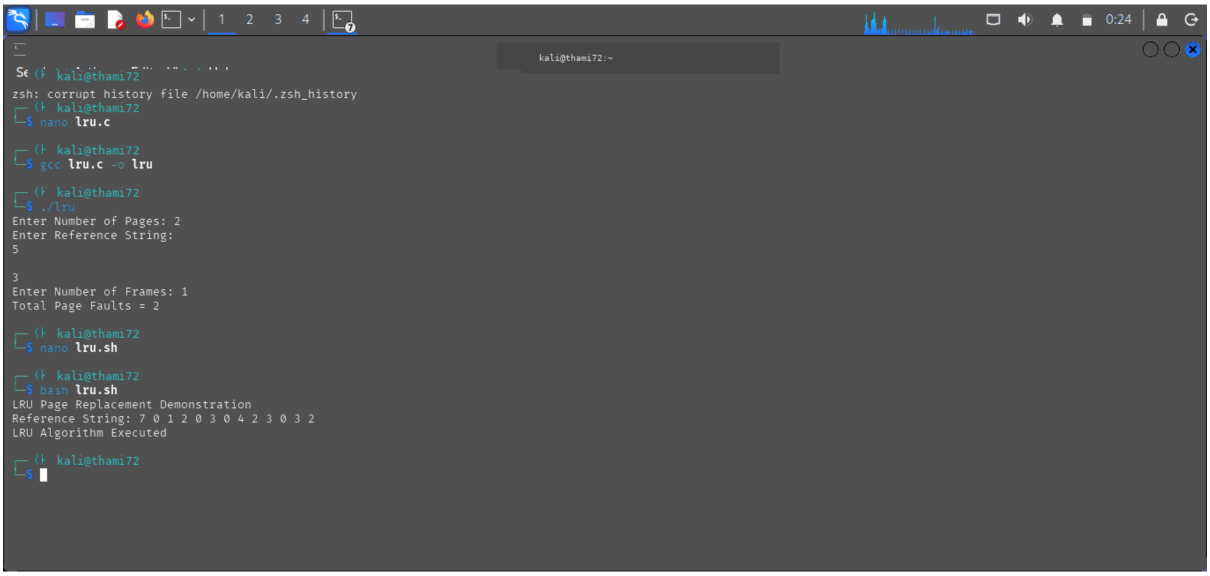
C PROGRAMMING :LRU

OUTPUT :



SHELL PRROGRAMMING : LRU

OUTPUT :



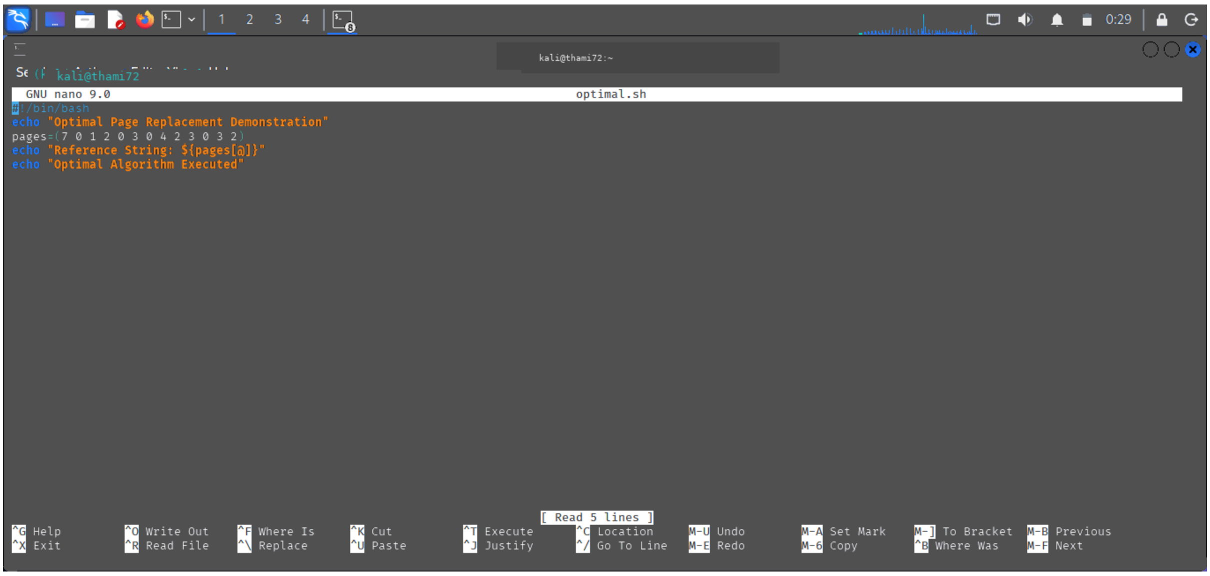
C PROGRAMMING :OPTIMAL PAGE

```
GNU nano 9.0                                optimal.c *
#include <stdio.h>
int main(){
int pages[50], frames[10], n, f, i, j, k, pos, fault = 0, found;
printf("Enter Number of Pages: ");scanf("%d", &n);
printf("Enter Reference String:\n");
for(i = 0; i < n; i++)
scanf("%d", &pages[i]);
printf("Enter Number of Frames: ");scanf("%d", &f);
for(i = 0; i < f; i++)
frames[i] =-1;
for(i = 0; i < n; i++){
found = 0;
for(j = 0; j < f; j++){
if(frames[j] == pages[i]){
found = 1;
break;}}
if(found == 0){
for(j = 0; j < f; j++){
int future = 999;
for(k = i + 1; k < n; k++){
if(frames[j] == pages[k]){
future = k;
break;}}
if(j == 0 || future > pos){
pos = future;}}
frames[0] = pages[i];
fault++;}}
printf("Total Page Faults = %d\n", fault);
return 0;}

^G Help      ^O Write Out  ^F Where Is  ^K Cut    ^T Execute  ^C Location    M-U Undo  M-A Set Mark  M-] To Bracket  M-B Previous
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy      ^B Where Was    M-F Next
```

OUTPUT :

```
zsh: corrupt history file /home/kali/.zsh_history
┌─(kali@thami72
└─$ nano optimal.c

┌─(kali@thami72
└─$ gcc optimal.c -o optimal

┌─(kali@thami72
└─$ ./optimal
Enter Number of Pages: 2
Enter Reference String:
2

3
Enter Number of Frames: 1
Total Page Faults = 2

┌─(kali@thami72
└─$ 
```

SHELL PROGRAMMING :



OUTPUT :